$$\mathrm{Tr}_{i,j}^{k+o}(f) = \left\langle \mathsf{st}, \mathbf{A}_f + \mathbf{B}_f^{[:,i+1:i+k+o]}\mathbf{C}_f^{[j+1:j+k+o,:]}, \mathbf{B}_f^{[:,1:i]}, \mathbf{C}_f^{[1:j,:]} \right\rangle$$